

CUSTOMER MILESTONE BRIEF

MS #43 - Oracle Source and Stored-Procedure Qualification

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.43.0
Date	2026-08-31
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #43 - Oracle Source and Stored-Procedure Qualification converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Upgrades the database proof with a genuine Oracle source adapter, SCN-bound resume, semantic-loss analysis, and bounded procedure translation.

Why this matters to a customer

Upgrades the database proof with a genuine Oracle source adapter, SCN-bound resume, semantic-loss analysis, and bounded procedure translation.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- Added a genuine Oracle semantic-core SourceAdapter with deterministic source discovery, privacy-preserving profiling, contract-bound extraction, content-bound SCN resume, and explicit fail-closed transaction capabilities.
- Added a 38-entry Oracle source compatibility ledger covering all 26 AUTHFRDS columns and Oracle number, date/time, empty-string, identifier, transaction, redo, DDL, sequence, procedure, package-state, and autonomous-transaction boundaries using all five governing classes.
- Added four bounded Oracle PL/SQL procedures and deterministic PL/pgSQL translations covering SELECT INTO, NO_DATA_FOUND, UPDATE and SQL%ROWCOUNT, decimal branches and application errors, and NVL/empty-string behavior.

- Added a 20-case stored-procedure corpus covering results, side effects, row counts, exception mapping, numeric thresholds, null/empty/space behavior, type failures, duplicate rows, and mutation boundaries.
- Added explicit fail-closed exclusions for dynamic SQL, autonomous transactions, package state, database links, and procedure-owned commits.
- Bound the source adapter, procedure conformance and five-class ledgers into eight independent Oracle-to-PostgreSQL gates; bounded development and the supported procedure subset pass while live source/target, redo, native procedure, full stored-logic, database-completion, and production claims remain false.

How it differs from earlier work

MS #43 builds on MS #42 (Integrated Pilot Qualification). The earlier milestone established its own bounded capability; this milestone adds oracle source and stored-procedure qualification while preserving the earlier evidence and its limits.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use oracle source and stored-procedure qualification as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- Bound the source adapter, procedure conformance and five-class ledgers into eight independent Oracle-to-PostgreSQL gates; bounded development and the supported procedure subset pass while live source/target, redo, native procedure, full stored-logic, database-completion, and production claims remain false.

Source of record

- CHANGELOG.md release section(s): 0.43.0
- README.md product and operating guidance

- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications
- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity

The underlying schemas, receipts, ledgers, tests, and policy decisions remain authoritative when greater detail is required.